

Roma Bulani

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Summary:

Senior Software Engineer specializing in Frontend Architecture with 5+ years of experience building scalable, high-performance web applications. Expertise in React, TypeScript, Micro Frontends, GraphQL, and modern UI systems. Led platform modernization initiatives, content authoring systems, and interactive learning experiences that improved developer productivity, authoring efficiency, and end-user engagement.

Skills:

- **Frontend Development:** React, Next.js, TypeScript, JavaScript, GraphQL, Apollo Client, Module Federation, Micro Frontends, Zustand, Redux Toolkit, Tailwind CSS, Localization (i18n), RTL Support
- **Backend Development:** Node.js, Express.js, MongoDB, Java, Spring Boot, Python
- **Testing:** Jest, React Testing Library
- **CMS & Content Platforms:** Contentful, Cloudinary
- **Build & Tooling:** Rollup, Rspack, Git

Work Experience:

Frontend Engineer L3 | [Alef Education](#) | UAE (Remote)

March 2025 - Present

- Led the migration of the learning player from a standalone **package-based architecture to a Micro Frontend (MFE)**, improving scalability, deployment flexibility, and integration across teacher and student platforms.
- Led cross-team collaboration to integrate the learning player and related platform features across teacher and student journeys, proactively driving alignment, resolving dependencies, and ensuring seamless end-to-end user experiences.
- Played a key role in developing the content authoring/editor platform, enabling curriculum authors to create, configure, and manage interactive learning content, contributing to a **50% improvement in authoring efficiency**.
- Designed and implemented a **micro-frontend architecture** using React, TypeScript, and Module Federation, enabling independent development and deployment for the editor app.
- Designed and implemented **advanced authoring capabilities on top of a TipTap-based editor**, including dynamic variables, fill-in-the-blank generation, dependency management, and content validation workflows for interactive learning experiences.
- Built and maintained a slide-based learning player capable of rendering multiple interactive widgets per slide, supporting multiple player modes, learner progress tracking, state persistence, resume functionality, localization using Lokalise, and full RTL support for Arabic language experiences.
- Built and **enhanced interactive educational widgets**, including Multiple Choice Questions, Fill in the Blanks, Matching Activities, Readers with Highlights & Notes, and other learning components used across the platform.
- Designed and implemented a widget communication and event framework to synchronize state and interactions between editor and player environments.
- Integrated GraphQL APIs using Apollo Client and improved platform performance through optimized state management (Zustand), rendering strategies, reusable abstractions, custom hooks, asset pickers, and shared UI components that accelerated development across multiple projects.

Senior Software Engineer | Cimpress ([Vista](#)) | Mumbai (Remote), India

July 2023 - March 2025

- Designed and developed reusable Smart Component libraries distributed as npm packages, **reducing frontend implementation effort by 90%** and driving consistency across multiple micro-frontends.
- Developed 3 Contentful Apps, streamlining content authoring and reducing errors by **50%** with preview options.
- Developed Social Feed Generation end-to-end, which compiles all the required data of products and sends it to the remote server of the vendor, this feed is used for tagging the social media posts with the actual products on the website. Further, I automated this process on a daily basis so as to reduce the errors caused by stale data.
- Developed the Data Resolvers and Resolver Gatsby Plugins as npm packages, which are used by different MFEs to get the resolved data which can then be passed to our Smart Components.

Software Engineer | Graphy By Unacademy | Bangalore (Remote), India**May 2022 - June 2023**

- Improved Lighthouse Score for [NextLevel](#), boosting SEO by **10%** and performance by **15%** for public pages. Enhanced mobile responsiveness and optimized public pages like Job Details and User Profile.
- Contributed to **25+** features for Graphy Repo, launched on [Product Hunt's](#) top 10 products. Enhanced UI/UX with visualizations, resulting in **10x** increase in viewers and engagement from 75 to 755 viewers per day. Implemented SEO optimization for **87%** organic traffic boost.
- Designed and developed analytics and revenue tracking dashboards using Recharts, enabling creators to visualize business metrics and performance insights. Also contributed to microtransactions, authentication flows, and user profile enhancements for Graphy 2.0.
- Worked on **10+** features for Graphy Link, a platform for scheduling 1:1 meetings. Notable features include Zoom Integration and meeting scheduling flow.

Application Development Analyst | Accenture | Mumbai (Remote), India**January 2021 - February 2022**

- Achieved Professional Training in JAVA, Spring MVC, and Spring Boot.
- Worked on Spring Boot Applications to fetch and validate the data from API and store it in SQL Database.
- Single-Handedly worked on migrating **30+** Unix Jobs from Oracle 12c to 19c during Database Upgrade.

Education:**Bachelor of Engineering (BE) in Computer Engineering** | University of Mumbai, India**2016 - 2020**

GPA: 9.4 / 10

Personal Projects:**[Bakin Lane](#)**, E-Commerce Application for cakes and muffins | [Source Code](#)

Full-stack e-commerce platform built with React, Express, MongoDB, Razorpay, and Cloudinary featuring JWT authentication, search, pagination, and payment integration.

[Bakin Gram](#), Social Media Application | [Source Code](#)

Social media platform built using React, Redux Toolkit, and Express with infinite scrolling, media uploads, and real-time user discovery.

Volunteer Experience:**Mentor** | [neoG Camp](#) | Mumbai (Remote), India**February 2023 - October 2023**

- Guided over 50 students in their Javascript and React interview preparation journey.
- Conducted live code reviews to promote better coding practices and resolving technical doubts and tried to motivate and inspire students to continuously improve and achieve their full potential.